THE CATEGORY OF LOCALLY QUASI-COMPACT HAUSDORFF SPACES HAS FIBRE PRODUCTS, A FINAL OBJECT AND QUASI-COMPACT FIBRE PRODUCTS OF QUASI-COMPACT OBJECTS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof is written out in full: the final object and Hausdorffness, the quasi-compact case through Tychonoff and closedness, and the general case through quasi-compact neighbourhoods.

Editorial note. The label lemma-LC-basic names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. The statement and proof are copied from the pinned chapter file; the five cited inputs are kept as external pointers with their canonical Stacks tags.

Allowed standard prerequisites (editorial).

- Hausdorffness of products and subspaces. Topology, the section on Hausdorff spaces, giving the Hausdorff property of X x Y and of its subspaces. Supplies the Hausdorff property of the fibre product as a subspace of the product.
- Tychonoff and closedness of the fibre product. Topology, Tychonoff's theorem and the lemma stating that the fibre product is closed in the product. Supplies quasi-compactness of the fibre product of quasi-compact objects.
- Criterion for being locally quasi-compact. Topology, the lemma characterising locally quasi-compact Hausdorff spaces through quasi-compact neighbourhoods. Supplies the final step producing a quasi-compact neighbourhood of an arbitrary point.

Source reference. Original lemma 005C: sources/stacks-chow/tags.txt, source lines 204–204. Prerequisite pointer invoked at proof line(s) 7494; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 08ZH: sources/stacks-chow/tags.txt, source lines 11375–11375. Prerequisite pointer invoked at proof line(s) 7492; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 08ZR: sources/stacks-chow/tags.txt, source lines 11384–11384. Prerequisite pointer invoked at proof line(s) 7502; it names a standard

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fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original section 08ZD: sources/stacks-chow/tags.txt, source lines 11371–11371. Prerequisite pointer invoked at proof line(s) 7486; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original theorem 08ZU: sources/stacks-chow/tags.txt, source lines 11387–11387. Prerequisite pointer invoked at proof line(s) 7490; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *The category LC has fibre products and a final object and hence has arbitrary finite limits. Given morphisms $X \to Z$ and $Y \to Z$ in LC with X and Y quasi-compact, then $X \times_Z Y$ is quasi-compact.*

Proof. The final object is the singleton space. Given morphisms $X \to Z$ and $Y \to Z$ of LC the fibre product $X \times_Z Y$ is a subspace of $X \times Y$. Hence $X \times_Z Y$ is Hausdorff as $X \times Y$ is Hausdorff by Topology, Section 08ZD.

If X and Y are quasi-compact, then $X \times Y$ is quasi-compact by Topology, Theorem 08ZU. Since $X \times_Z Y$ is a closed subset of $X \times Y$ (Topology, Lemma 08ZH) we find that $X \times_Z Y$ is quasi-compact by Topology, Lemma 005C.

Finally, returning to the general case, if $x \in X$ and $y \in Y$ we can pick quasi-compact neighbourhoods $x \in E \subset X$ and $y \in F \subset Y$ and we find that $E \times_Z F$ is a quasi-compact neighbourhood of (x, y) by the result above. Thus $X \times_Z Y$ is an object of LC by Topology, Lemma 08ZR. □